

Sasuni Wijerathne

Full-Stack Engineer | Mobile Developer | AI/ML Engineer

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PROFESSIONAL SUMMARY

Results-driven Full-Stack Software Engineer, Mobile Application Developer, and AI/ML Engineer with hands-on experience designing, developing, testing, and deploying production-ready web, mobile, and AI-powered applications. Proficient in React.js, Next.js, React Native, Node.js, Java, Python, and Spring Boot. Experienced in machine learning model development, data analysis, and AI integration using OpenAI API, Google Generative AI. Skilled in cloud infrastructure, containerisation (Docker, Kubernetes), CI/CD pipelines, and DevOps automation. Strong communicator with Agile/Scrum experience, delivering scalable solutions from concept to production.

SKILLS

Languages Java, Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
Frontend React.js, Next.js, Redux, Tailwind CSS, Bootstrap, Responsive Web Design
Mobile React Native, Expo, Android Development, NativeWind, Cross-Platform Dev
Backend Spring Boot, Node.js, Express.js, RESTful APIs, Microservices, WebSockets
Databases MySQL, MongoDB, Firebase, Query Optimisation
AI & ML Pandas, NumPy, OpenAI API, Google Generative AI
DevOps & Cloud Docker, Kubernetes, GitHub Actions, CI/CD, AWS (EC2, S3), Azure, Nginx
Security Spring Security, JWT, OAuth2, Data Encryption
Tools Git, Swagger, Postman, Figma, VS Code, IntelliJ IDEA, FFmpeg, Android Studio
Soft Skills Problem Solving, Team Collaboration, Communication, Leadership, Agile/Scrum

TECHNICAL EXPERIENCE - PROJECTS

PropertyPulse – AI-Powered Real Estate Platform <i>Full-Stack Software Engineer</i>	Dec 2025 – Jan 2026 Colombo, Sri Lanka
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- Architected and developed a full-stack AI-powered real estate web application managing 100+ property listings with intelligent market analysis, automated price prediction, and AI-generated descriptions using OpenAI API and Google Generative AI.
- Engineered role-based access control (RBAC) and integrated Stripe payment gateway, enabling secure multi-tier transactions and reducing unauthorised access incidents by 100%.
- Designed scalable RESTful APIs with Node.js and Express.js; optimised MongoDB queries improving data retrieval speed by 35%.
- Stack:** React.js, TypeScript, Node.js, Express.js, MongoDB, OpenAI API, Google Generative AI, Stripe, Tailwind CSS, Docker, GitHub Actions

Strategic Patient Risk Stratification System <i>Machine Learning Engineer / Data Scientist</i>	Dec 2025 – Feb 2026 Colombo, Sri Lanka
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- Designed, trained, and validated a supervised ML classification model achieving 85% accuracy in identifying high-risk diabetic patients for 30-day hospital readmissions.
- Developed the Vitality Complexity Index (VCI) scoring algorithm integrating clinical data, demographics, and historical records, improving risk stratification precision by 30%.
- Produced 5+ interactive dashboards using Matplotlib and Seaborn, reducing stakeholder decision-making time by 40% and supporting treatment planning for 200+ patient records.
- Stack:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, NLP, Predictive Modelling

- Architected a scalable backend supporting tiered membership plans, secure authentication, and AI-powered content generation via OpenAI API, reducing content creation time by 60%.
- Implemented real-time bidirectional communication via WebSockets; integrated OAuth2 social login and full REST API documentation via Swagger.
- Enforced security with Spring Security, JWT, and RBAC; achieved 90%+ test coverage, reducing post-deployment bugs by 50%.
- **Stack:** Java, Spring Boot, Spring Security, MySQL, JWT, OAuth2, WebSockets, Swagger, Microservices

- Developed an end-to-end AI video generation pipeline integrating text-to-speech via ElevenLabs API with automated script generation and media assembly, reducing production time by 70%.
- Engineered FFmpeg-based media workflows automating video rendering, thumbnail generation, and file optimisation, processing 100+ videos per session.
- Built and deployed a Flask REST API backend maintaining 99% uptime.
- **Stack:** Python, Flask, FFmpeg, ElevenLabs API, HTML5, CSS3, JavaScript, Docker

EDUCATION

Higher National Diploma (HND) in Software Engineering Institute of Software Engineering (IJSE), Colombo, Sri Lanka	2024 – Present
Certified AI & ML Engineer Program (CAME) Institute of Software Engineering (IJSE), Colombo, Sri Lanka	2025 – Present
G.C.E. Advanced Level – Mathematics Stream (IT) St. Paul’s Milagiriya, Colombo 03, Sri Lanka	2023

ACTIVITIES & ACHIEVEMENTS

• Hackathon Winner – 1st Place , GENESYS Hackathon, Team CodeHub, IJSE (20+ teams)	2024
• Secretary , IJSE Student Committee – Managed 15+ events, led 60+ students	2026
• Vice Secretary , IJSE Student Committee	2025
• Competitive Programmer , CodeRally 6.0 Hackathon, IIT	2025
• Group Project – AutoTrade Vehicle Bidding Platform, Circle Edge, IJSE	2025
• Technical Writer – Published 10+ articles on Medium	2024 – Present

REFERENCES

Udara San Founder, Director & CEO SANS Technologies +94 78 840 9788 sans@sans-technologies.com	Dulan Herath Software Engineer – AI Sysco Labs +94 71 761 5558 dulan.herath@sysco.com
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